order36 RESULTS: RG33 (leakage as a span projection)
PREREG sha256 3b7fc650d363d1d0c904276483a25879c47eb7cff2dff91086c9134d80146f9e (hashed before numerics).
Span = column span of the companion Gauss means G = A0/I (ambient rank 36 = dim V; the 35-component law vector is a separate notion).
Grid handling (as preregistered)
- P7 (order20 grid) is a trivial control: rank G(P7) = 7, so every column lies in the span. Leftover is 6.4e−16 as expected; non-discriminating.
- Primary grid G48: the 48 lowest-complexity points of XS. rank G(G48) = 36, as required. Results are reported globally and per point.
Controls
D0 (L_B split reconstructs C2b(−1/x)): 1.9e−14 on G48, PASS.
RG33: FAIL (Category B)
quantity (G48)	value
r_meas (κ̂ 0.99971 − 0.0043i)	0.01899
r_pred = ‖(I−P_G)S_out‖/‖S‖, no fit	0.3590
relative difference (PASS ≤ 1e−6)	17.9
‖(I−P_G)S‖/‖S‖	0.3678
‖(I−P_G)X‖/‖X‖	0.3677
‖(I−P_G)S_in‖/‖S‖	0.074
Spearman(per-point r, per-point leftover)	0.18
Per-point r: median 0.0187, range 0.0093 (x = 5/8) to 0.139 (x = 1). The worst points are 1, 2, ½, 3, the smallest denominators; per-point leftover median 0.46. Prediction FAIL (blind): correct.
Reading
- S and the law direction X have the same out-of-span fraction (0.3678 vs 0.3677). Both live largely outside the span of the plain companion Gauss means, and their outside parts are aligned. That is why the near-law survives while the projection leftover is large.
- The plain-Gauss-mean column span is therefore the wrong reference space for the Bernoulli sum C2b. Leakage "outside the companion Gauss span" does not measure the law miss.
- The near-law quality is strongly grid-dependent: r = 0.0585 on P7, 0.019 on G48 and 0.00305 on the full grid. It improves as larger-denominator points are added, and the miss concentrates at x ∈ {1, 2, ½, 3}.
Category
A: none new. v11: no. Ξ: not identified. E1–E3 OPEN.

order36 NOTES
- Two leakage definitions have now failed: by characteristic (order35) and by plain-Gauss span (order36). Both S and X sit outside the plain span by the same 37%.
- The natural reference space for C2b is the span of the Bernoulli-sum columns {C2b_l} themselves, together with the order20 RG7 joint-rank picture. That is where the fixed-M law lives.
- Grid dependence is now the most concrete pattern: the miss concentrates at small-denominator x and shrinks as the grid fills in (0.0585 → 0.019 → 0.003).
- Open next step (order37, not started): preregister RG34, a per-point law miss as a function of the denominator of x. Hypothesis to fix before testing: per-point r(x) for C2b decays like c·den(x)^{−p}, with p and c fixed by a stated rule (e.g. from L_B, where N′ = 2|num(x)| and the aliasing weights c′_k ∝ 1/k² suggest p = 2). Test on the full 216 grid using stored arrays only (cheap). Thresholds fixed in advance.
- Naming: RG* only, no labels L1–L7. Geometric only; MAPS_CANDIDATES untouched; no Resonant objects; no bridge claim; CORE_FREEZE. nice, OMP_NUM_THREADS=1, 3 GB, ~15 min; no push or sends; new files only.